CS471 – Lesson 7 Solutions Key

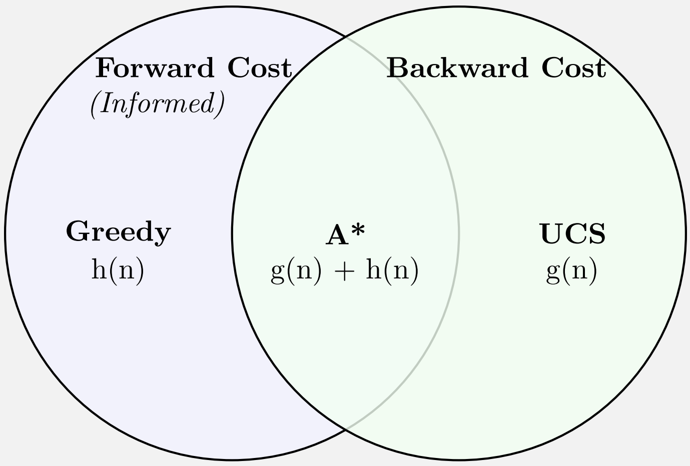
Heuristics, Admissibility, and Consistency

Part 1: Guided In-Class Worksheet (Instructor-Led)

1. Search Algorithms Venn Diagram

Draw a Venn diagram showing the relationship between the following searches: **DFS, BFS, UCS, Greedy, A***. Organize them by cost (forward, backward, both, none). Note which ones are considered"informed" searches.

Solution

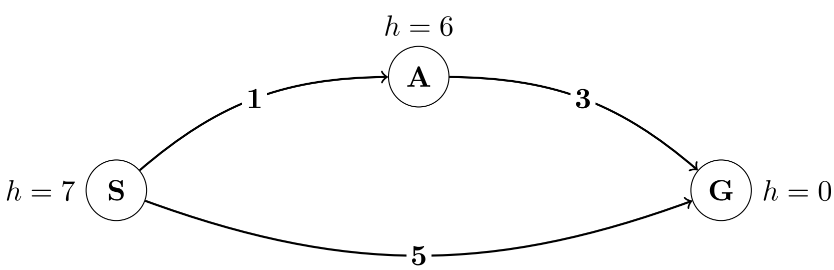


BFS, DFS
(No Cost Evaluated)

Note: Anything utilizing a forward cost estimate (a heuristic of the goal) is classified as an informed search.

2. Why A* Breaks

Evaluate the graph below. The true optimal solution is $S \to A \to G$ with a cost of 4. However, A* breaks on this graph and will select the suboptimal path $S \to G$ (cost 5).



Solution

Why does A* break here?
A* evaluates $f(n) = g(n) + h(n)$.

- Direct path to G: $f(G) = 5 + 0 = 5$
- Path to A: $f(A) = 1 + 6 = 7$

Because the heuristic at node A is so pessimistic (6), it artificially inflates the cost of that path. A* sees $5 < 7$, dequeues G, and terminates early, locking in the suboptimal path.

What would the heuristic at node A need to be to fix this?
To ensure A* explores node A before terminating at G, $f(A)$ must be less than or equal to $f(G)$. Therefore, $1 + h(A) \leq 5$, meaning $h(A)$ would have to be **at most 4**. *(Strictly speaking, for it to be completely admissible, it must be ≤ 3, which is the true remaining cost from A to G).*

3. Core Definitions

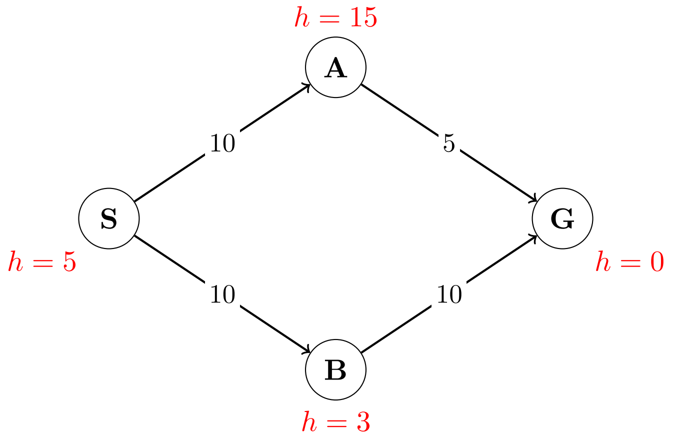
Solution

1. **Admissibility:** For all nodes n, $0 \leq h(n) \leq h^*(n)$, where $h^*(n)$ is defined as the true, exact cost from n to a goal state. (The heuristic never overestimates the cost).
2. **Consistency:** For all nodes A and B, $h(A) - h(B) \leq \text{cost}(A \to B)$. (The triangle inequality; the heuristic estimate cannot drop faster than the true cost to move between two nodes).
3. **Dominance:** Heuristic h_a dominates heuristic h_b if and only if for all nodes n, $h_a(n) \geq h_b(n)$. Dominating heuristics lead to fewer nodes being expanded in A* search because they are closer to the true cost without going over.

Part 2: Student Independent Practice Solutions

4. Evaluating Heuristic Properties

Graph 1:



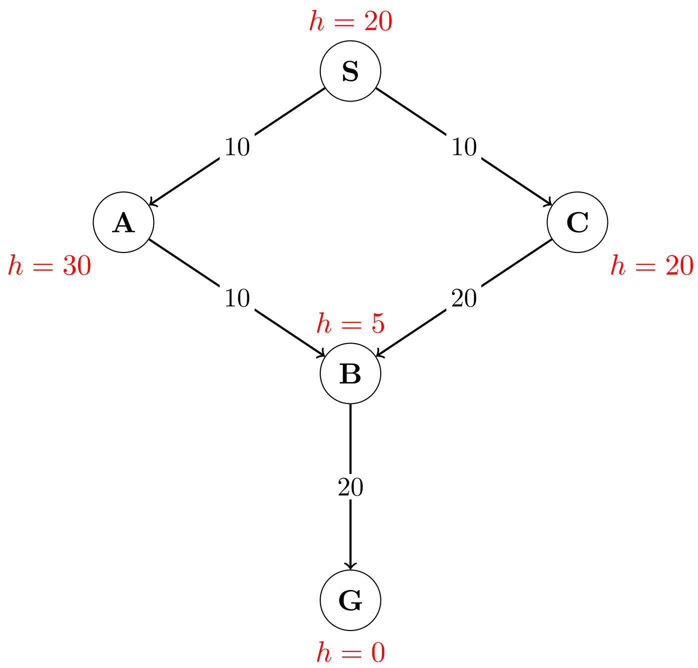
Solution

Violations: Both (Admissibility and Consistency)

- **Admissibility Violated:** $h(A) \leq h^*(A)$? No. $15 \not\leq 5$.
- **Consistency Violated:** $h(A) - h(G) \leq \text{cost}(A \to G)$? No. $(15 - 0) \not\leq 5$.

Note: If a heuristic is not admissible, it mathematically cannot be consistent.

Graph 2:



Solution

Violations: Consistency Only

- **Admissibility Valid:** We can confirm $h(n) \leq h^*(n)$ for all n in the graph. The heuristic is optimistic everywhere.

- **Consistency Violated:** $h(A) - h(B) \leq \text{cost}(A \rightarrow B)$? No. $(30 - 5) \not\leq 10$. The heuristic estimate "shrank" by 25 units, even though we only moved 10 units.

5. Combining Admissible Heuristics

Solution

1. $a(n) + b(n) + c(n)$: **No.** The sum could easily exceed the true cost $h^*(n)$, making it an overestimate (inadmissible).

2. $\min\{a(n), b(n), c(n)\}$: **Yes.** The minimum of multiple admissible heuristics will always be $\leq h^*(n)$ and non-negative.

3. $\max\{a(n), b(n), c(n)\}$: **Yes.** The maximum is still $\leq h^*(n)$ and non-negative. Furthermore, this approach is optimal because it creates a **dominant** heuristic over a, b, and c individually.

4. $\frac{a(n)+b(n)+c(n)}{3}$: **Yes.** The average of valid estimates is guaranteed to be $\leq h^*(n)$ and non-negative.

5. 0: **Yes.** 0 is always $\leq h^*(n)$ and non-negative. However, it is an extremely weak heuristic and is dominated by all other heuristics (using $h = 0$ essentially turns A* back into Uniform Cost Search).